

AWS Solutions Architect Fundamentals

The non-GenAI infrastructure services that appear throughout this exam — what they do, and the details that decide questions.

Why this document exists: roughly a third of the GenAI Professional exam rests on ordinary AWS architecture — networking, identity, integration, storage. Task 2.3 in particular is almost entirely Solutions Architect content with a GenAI use case bolted on. If these fundamentals are solid, those questions become straightforward.

Green boxes mark **where each service showed up in the GenAI course**, so you can connect the fundamental to its application.

1. Networking & Content Delivery

Traffic routing — the four options

Service	What it does	Choose when
Route 53	DNS. Routing policies: simple, weighted, latency, failover, geolocation, geoproximity, multi-value.	DNS-level routing is enough and TTL-based failover delay is acceptable
Global Accelerator	Two static anycast IPs announced from all edge locations. Traffic enters the nearest edge, then travels the AWS backbone to the closest healthy regional endpoint.	Fast failover (seconds, not TTL), non-cacheable traffic, static IPs needed for allowlists
CloudFront	CDN — caches content at edge locations	Content is cacheable
ELB (ALB / NLB)	Distributes traffic within a region. ALB = layer 7 (HTTP), NLB = layer 4 (TCP, ultra-low latency, static IP per AZ)	Regional load distribution

Global Accelerator, precisely: anycast routing gets the packet to the nearest edge — **GA is not involved in that part**. GA takes over *at* the edge and decides which region receives it. The two static IPs are for **redundancy across network zones**, not one per region. IPs never change, which is why partner firewalls can allowlist them.

In the course: Task 1.1 — routing global bank users, with the caveat that GA has no native "EU users stay in EU" setting (use traffic dials at 0% or separate accelerators).

VPC fundamentals

Component	Key points
Subnets	Public (route to internet gateway) vs private (no direct internet). Each subnet lives in one AZ.
Security groups	Stateful , instance-level, allow rules only . Return traffic is automatic.
Network ACLs	Stateless , subnet-level, allow <i>and</i> deny rules, evaluated in numbered order. Return traffic needs its own rule.
NAT Gateway	Lets private subnets reach the internet outbound only. Managed, per-AZ, charged hourly + per GB.
Internet Gateway	Attaches to the VPC; enables two-way internet for public subnets

The classic exam pair: security group = stateful, allow-only, on the resource. NACL = stateless, allow+deny, on the subnet. **Egress rules matter for AI workloads** — restricting outbound traffic is what prevents a compromised component from exfiltrating data.

VPC endpoints and PrivateLink

Type	Details
Interface endpoint	Creates an ENI in your subnet with a private IP. Works for most services (Bedrock, Comprehend, SageMaker, KMS, SQS...). Supports security groups and endpoint policies . Charged hourly + per GB.
Gateway endpoint	A route table entry , not an ENI. Only S3 and DynamoDB . Free.
PrivateLink	The underlying technology for interface endpoints; also lets you expose your own service privately to other VPCs

In the course: Task 3.2 — keeping all healthcare AI traffic inside the VPC; Task 2.3 — endpoint policies restricting *which Bedrock operations* the enterprise network may invoke.

Hybrid and multi-VPC connectivity

Service	Purpose and key detail
Direct Connect	Dedicated physical link to AWS, bypassing the internet. Consistent bandwidth and latency. BGP routing policies can prioritise some traffic over others.
Site-to-Site VPN	Encrypted tunnel over the internet. Cheaper and faster to set up than Direct Connect; less consistent.
Transit Gateway	Hub connecting many VPCs, VPNs and Direct Connect. Route tables provide domain isolation (dev / test / prod). Black hole routes deliberately drop traffic.
VPC Peering	One-to-one, non-transitive. Fine for a few VPCs; Transit Gateway scales better.

Edge locations — the three "not a region" options

Service	Where it sits	Deciding factor
Outposts	AWS racks in your data centre	Compliance — data legally cannot leave the premises
Local Zones	AWS infrastructure in major metro areas	Distance — users far from the nearest region
Wavelength	Inside a telecom carrier's 5G network	Mobile — traffic never touches the internet

AWS WAF and Shield

- WAF** — layer 7 filtering on CloudFront, ALB, API Gateway, AppSync. Rule types: rate-based, geo-match, SQL injection, XSS, IP sets, managed rule

groups.

- **Shield Standard** — free DDoS protection; **Shield Advanced** — paid, with response team and cost protection.

In the course: rate-based rules matter unusually much for GenAI, because **every request costs money** — abuse is a billing attack, not just a load problem.

2. Compute

Lambda

Concept	Detail
Limits	15-minute maximum execution; 10 GB memory ceiling; 512 MB-10 GB ephemeral storage in <code>/tmp</code>
Reserved concurrency	A cap — this function cannot exceed N concurrent executions, protecting the rest of the account
Provisioned concurrency	Pre-warmed instances — no cold starts, but you pay while idle. Only worth it with predictable traffic.
Function URLs	Direct HTTPS endpoint. Supports response streaming — API Gateway does not.
Layers	Shared dependencies across functions
In a VPC	Gets an ENI; needs a NAT gateway or VPC endpoints for outbound access

Containers and instances

Service	Choose when
ECS	Container orchestration, AWS-native. Fargate = serverless containers; EC2 launch type = you manage the instances.
EKS	Managed Kubernetes — choose when you need Kubernetes specifically
EC2	Full control. Instance families: M general, C compute, R memory, G/P GPU, Inf Inferentia (inference), Trn Trainium (training).
Auto Scaling	Target tracking, step, scheduled, and predictive (uses historical patterns to scale <i>before</i> the spike)

Reactive scaling arrives late. By the time metrics trigger and instances warm up, the spike has passed. Predictive scaling and scheduled scaling exist for known patterns — market open, morning rounds, shift changes.

In the course: ECS hosts heavy MCP servers (Task 2.1); Lambda hosts light ones. Inferentia appears in Task 4.2 as the cost-efficient inference chip.

3. Storage & Databases

Amazon S3

Feature	Detail
Storage classes	Standard → Intelligent-Tiering → Standard-IA → One Zone-IA → Glacier Instant / Flexible / Deep Archive
Lifecycle policies	Automatic transition between classes and expiration . In regulated settings this is a retention control , not just cost management.
Versioning	Keeps every version; needed for reproducibility and accidental-delete protection
Encryption	SSE-S3 (AWS-managed), SSE-KMS (your key, auditable), SSE-C (you supply the key), client-side
Object metadata / tags	Metadata capped around 2 KB; only 10 tags per object — supplement with a database for richer metadata
Event notifications	Trigger Lambda, SQS or SNS on object create/delete

Databases — choosing

Service	Nature	Choose when
DynamoDB	NoSQL key-value, single-digit ms	Known access patterns, high scale. TTL for automatic expiry; Streams for change capture; GSI for alternative query patterns.
RDS / Aurora	Relational	Joins, transactions, complex queries. Aurora = AWS-built, faster, auto-scaling storage.
ElastiCache	In-memory (Redis / Memcached)	Sub-millisecond reads on hot data
OpenSearch	Search and analytics	Full-text search, log analytics, vector search
Neptune	Graph	Relationship traversal
Timestream	Time-series	Metrics over time, trend analysis

DynamoDB design points that appear repeatedly: partition key determines distribution (avoid hot partitions); **GSI** enables queries on non-key attributes; **TTL** deletes items automatically — which is how a data-retention policy becomes self-enforcing rather than a script someone forgets to run.

In the course: DynamoDB stores conversation history with 90-day TTL (Task 1.6), agent memory tiers (2.1), idempotency keys (2.3). ElastiCache serves as the sub-ms memory tier and response cache.

4. Application Integration

Messaging — the five-way comparison

Service	Model	Choose when
SQS	Queue, pull-based	Decoupling, buffering, smoothing spikes. Visibility timeout must exceed processing time or a second consumer picks up the same message.
SQS FIFO	Ordered queue	Order matters. Message group ID preserves order within a group while allowing parallelism across groups.
SNS	Pub/sub, push	Fan-out to many subscribers, no content filtering (though filter policies exist)
EventBridge	Event bus with rules	Content-based routing , SaaS sources, schema registry, input transformers, archive and replay
Kinesis Data Streams	Ordered stream, replayable	Real-time analytics, multiple consumers reading the same data, retention window
Amazon MQ	Managed ActiveMQ / RabbitMQ	Only when migrating a system that already speaks JMS/AMQP/MQTT

Dead-letter queues apply to SQS, SNS and Lambda. After N failed attempts the message parks in the DLQ instead of being lost — essential wherever events are audit-relevant.

Orchestration and APIs

Service	Key points
Step Functions	State machine. States: Task, Choice, Parallel, Map, Wait, Pass, Succeed, Fail. Map applies the same work to many items and takes <code>MaxConcurrency</code> . Task tokens pause execution for a human. Standard (long-running, exactly-once) vs Express (high-volume, at-least-once).
API Gateway	REST (features, usage plans, caching) vs HTTP (cheaper, faster, fewer features) vs WebSocket. ~30s integration timeout ; usage plans + API keys; request validators; mapping templates; custom authorizers; cannot stream responses .
AppSync	Managed GraphQL. Subscriptions over WebSocket ; field-level authorisation; VTL or JS resolvers; direct data source integration.
AWS Glue	Serverless ETL. Crawlers populate the Data Catalog ; job bookmarks give incremental processing.
AppFlow	Pre-built SaaS connectors (Salesforce, ServiceNow, Slack), bidirectional, filtered

5. Identity, Security & Governance

IAM

Concept	Detail
Policy types	Identity-based (on user/role/group), resource-based (on the resource — S3 bucket policy, KMS key policy), permission boundaries , SCPs , session policies
Evaluation logic	Explicit deny always wins . Otherwise: allow must exist somewhere and nothing may deny it. SCPs and boundaries act as filters on the maximum.
Condition keys	<code>aws:SourceIp</code> , <code>aws:CurrentTime</code> , <code>aws:SourceVpce</code> , <code>aws:RequestTag</code> , <code>aws:PrincipalTag</code>
Roles vs users	Roles are assumed and give temporary credentials — always preferred. Cross-account access uses role assumption with a trust policy.
RBAC vs ABAC	RBAC = permissions by role. ABAC = permissions from tags/attributes , so new cases don't need new policies.

IAM vs SCP — the distinction the exam leans on: IAM **grants** permission. An SCP sets a **ceiling** and never grants anything. If an SCP denies it, no IAM policy in that account can allow it — **not even for the account administrator**.

Identity services

Service	For
IAM Identity Center	Workforce — SSO for employees across accounts; federates with your corporate IdP
Cognito User Pools	Customer sign-up/sign-in; issues JWTs. Lambda triggers (pre-token generation, pre sign-up, post confirmation) let you enrich or intercept.
Cognito Identity Pools	Exchanges a token for temporary AWS credentials
Verified Permissions	Application-level authorisation using the Cedar policy language — "may this user act on <i>this record</i> ", which IAM cannot express

JWT validation is local. API Gateway and AppSync verify the token's signature against Cognito's cached public keys — no call to Cognito per request. Consequence: **a deleted user's token remains valid until it expires** (typically an hour).

Encryption and secrets

Service	Detail
KMS	Key management. Customer-managed keys give you the key policy and audit trail; AWS-managed keys are simpler. Multi-Region keys replicate key material. Grants give temporary fine-grained access without editing the policy. Automatic annual rotation.
Secrets Manager	Secrets with automatic rotation ; costs per secret
Parameter Store	Configuration and secrets; free standard tier; no built-in rotation
ACM	TLS certificates, free for AWS services, auto-renewing

Governance and detection

Service	Purpose
Organizations	Multi-account structure with OUs; the container for SCPs and consolidated billing
Control Tower	Landing zone with controls (formerly guardrails) — preventive (SCP-based) and detective (Config-based). Blocks non-compliant deployments <i>before</i> they happen.
CloudTrail	API call audit. Management events on by default; data events are OFF — enable them explicitly for object-level access logging. Multi-region trails; analyse with Athena.
AWS Config	Resource configuration history and compliance rules
Security Hub	Aggregates findings from GuardDuty, Macie, Inspector, Config into one view
GuardDuty	Threat detection from logs — unusual API activity, known-bad IPs
Macie	Discovers and classifies sensitive data sitting in S3
Audit Manager	Automates evidence collection against a compliance framework
RAM	Shares resources across accounts, with tag-based conditions
Lake Formation	Column, row and cell-level permissions inside data lake tables — a granularity IAM cannot reach

6. Observability & Operations

CloudWatch and X-Ray

Feature	Detail
Metrics	Standard (free, 5-min) vs detailed (1-min) vs custom metrics (you publish them)
Alarms	Static thresholds, anomaly detection (learns a baseline, understands seasonality), and composite alarms combining several conditions
Logs	Log groups with retention settings; Logs Insights for querying; metric filters turn log patterns into metrics
RUM	Real User Monitoring — client-side measurement of what the user actually experienced
Synthetics	Canaries that generate their own traffic on a schedule — detects problems before real users do
Evidently	Feature flags and A/B experiments with statistical analysis
X-Ray	Distributed tracing — segments, subsegments, annotations (indexed, filterable) and metadata. Sampling rules keep cost down: low rate for normal traffic, 100% for errors.

CloudWatch vs X-Ray: CloudWatch tells you *how slow* and *how many*. X-Ray tells you **where** the time went across services. Neither substitutes for the other, and troubleshooting usually needs both.

Deployment and IaC

Service	Detail
CloudFormation	IaC. DependsOn forces creation order; Conditions vary config per environment; custom resources run Lambda for anything unsupported; StackSets deploy across accounts and regions.
CDK	IaC in a programming language, synthesised to CloudFormation
CodePipeline	Orchestrates stages; supports manual approval actions
CodeBuild	Runs builds and tests; caching avoids re-downloading dependencies
CodeDeploy	Deployment strategies with healthy host thresholds and CloudWatch alarm-triggered rollback
Service Catalog	Approved product portfolios; launch constraints enforce security baselines and tagging on self-service deployments
AppConfig	Runtime configuration and feature flags — changes apply without redeploying. Deployment strategies (linear, canary, all-at-once) with automatic rollback on alarm.

Deployment strategies

Strategy	Behaviour
In-place / rolling	Replace instances gradually; no extra capacity, slower rollback
Blue/Green	Two full environments, switch at once. Instant rollback , double the cost during cutover.
Canary	Small percentage first, then increase
Linear	Fixed increments at fixed intervals
Shadow	New version processes real traffic but its output is discarded — zero user risk, double cost

Cost management

Service	Purpose
Cost Explorer	Analyse historical spend; group by tag, service, account
AWS Budgets	Proactive — set thresholds and alert or act on breach
Cost Allocation Tags	The mechanism that makes per-team and per-feature attribution possible
Compute Optimizer	Right-sizing recommendations from utilisation data

7. Well-Architected Framework

Pillar	Core question	GenAI-specific concerns
Operational Excellence	Can we run and improve it?	Prompt versioning, model evaluation pipelines, GenAIOps
Security	Is it protected?	Prompt injection, PII, Guardrails, per-model access
Reliability	Does it recover?	Throttling, fallback models, graceful degradation
Performance Efficiency	Are resources used well?	Model selection, caching, streaming
Cost Optimization	Are we spending wisely?	Token optimisation, right-sized models, batch, caching
Sustainability	Minimising impact	Smallest model that does the job

Reviews run through the **Well-Architected Tool**, with the **Generative AI Lens** selected.

The SA Distinctions Most Likely to Appear

Pair	The deciding factor
Security group vs NACL	Stateful/instance vs stateless/subnet; NACL can deny
Interface vs Gateway endpoint	Gateway is only S3 and DynamoDB, and free
Global Accelerator vs CloudFront	Cacheable content → CloudFront; otherwise GA
Global Accelerator vs Route 53	Failover speed — GA is seconds, Route 53 waits on TTL
SQS vs SNS vs EventBridge	Queue vs fan-out vs content-based routing
SQS vs Kinesis	Kinesis when you need ordering, replay, or multiple independent consumers
Transit Gateway vs Peering	Many VPCs, or transitive routing needed → TGW
IAM vs SCP	Grant vs ceiling; SCPs cannot be bypassed by account admins
Reserved vs provisioned concurrency	Cap vs pre-warmed (and paid-for) capacity
Secrets Manager vs Parameter Store	Automatic rotation needed → Secrets Manager
Blue/Green vs Canary vs Shadow	Everyone at once / a few users / nobody sees it
CloudWatch vs X-Ray	How slow vs where it was slow
Config vs CloudTrail	What the resource <i>is</i> vs who <i>did</i> something

Service limits and features change. Verify specific numbers against current AWS documentation before the exam.